MIL-E-15597E(MR)
24 November 1980
SUPERSEDING
MIL-E-155597D(MR)
25 September 1967

MILITARY SPECIFICATION

ELECTRODES, WELDING, COVERED, COATED ALUMINUM AND ALUMINUM ALLOY

This specification is approved for use by the Army Materials and Mechanics Research Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers coated or covered metal arc welding electrodes for welding aluminum and aluminum alloys.

1.2 Classification. The electrodes shall be of the following types and sizes, shown in Table I as specified (see 6.2):

Table I. Types, sizes and positions

Type	AWS Designation	Sizes, in (mm)	Welding Positions 1/ Groove weld	Fillet weld
MIL-1100	E1100	3/32 (2.4), 1/8 (3.2),	F, V	F, V, H
		5/32 (4.0), 3/16 (4.8)	F, V	F, V, H
		1/4 (6.4)	F	F
MIL-3003	E3003	3/32 (2.4), 1/8 (3.2),	F, V	F, V, H
		5/32 (4.0), 3/16 (4.8)	F, V	F, V, H
		1/4 (6.4)	F	F
MIL-4043	E4043	3/32 (2.4), 1/8 (3.2),	F, V	F, V, H
		5/32 (4.0), 3/16 (4.8)	F, V	F, V, H
		1/4 (6.4)	F	F

1/ F = flat, V = verticle, H = horizontal

1.2.1 Length. Unless otherwise specified (see 6.2) welding electrodes shall be 14 inches (355.6 mm) ± 1/4 inch (6.4 mm) in length.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Army Materials & Mechanics Research Center, Watertown, MA 02172 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2. The following documents of the issue in effect on date of invitations for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-W-10430 - Welding Rods and Electrodes, Preparation for Delivery of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN WELDING SOCIETY (AWS)

A3.0 - Welding Terms and Definitions
A5.3 - Aluminum & Aluminum Alloy Arc-Welding Electrodes
Z49.1 - Safety in Welding and Cutting

(Applications for copies should be addressed to the American Welding Society, 2501 N. W. 7th Street, Miami, FL 33125.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. The electrodes furnished under this specification shall be a product(s) which has been inspected and has passed the first article inspection specified herein or has been accepted on waiver of first article inspection requirements. The contractor shall furnish a sample unit(s) for first article inspection and approval when specified (see 4.5, 6.2, and 6.3).

3.2 Definitions. For the purpose of this specification the welding terms and definitions contained in AWS A3.0 shall apply.

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3.3 Material. Types MIL-1100, MIL-3003 and MIL-4043 electrodes shall conform to all the chemical and mechanical requirements specified for E1100, E3003 and E4043 in AWS A5.3 respectively and the requirements of this specification.

3.3.1 Electrode end. The core wire, of the arc end of the electrode, may be completely coated with the covering material or exposed to no more than 1/8 inch (3 mm).

3.3.2 Storage life. Coverings shall be such that electrodes will meet the requirements of this specification after being stored as specified in the contract or order (see 6.2 & 6.4).

3.3.3 Fumes. Fumes from the burning coverings of electrodes can be injurious to personnel. Adequate ventilation should, in accordance with the recommendations of AWS Z49.1, be provided.

3.3.4 Slag removal. Slag deposited by the coverings shall be readily removable from the cooled weld deposits with the use of had tools (not air-or power-operated tools).

3.4 Deposit positions. Electrodes shall be capable of depositing satisfactory welds of all types in the positions specified in Table I (see AWS A3.0 and AWS 5.3).

3.5 Electrode identification. Electrodes shall be in accordance with AWS 5.3.

3.6 Workmanship. Electrodes shall be manufactured in accordance with high-grade commercial practice. The electrodes shall be of uniform quality, clean, sound and smooth. The electrode covering shall be free from scabs, scale, cracks, blisters, pock marks, bruises or other surface defects. The electrodes shall be free from any substances or characteristics which could adversely affect the operation of the electrodes or cause unsound weld metal.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. A lot shall consist of all electrodes of the same heat or core wire chemical analysis, type, and size, covered with the same coating material, packaged as specified in the contract and offered for inspection at one time.

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4.3 Sampling.

4.3.1 Sampling for first article inspection. First article test samples shall be obtained from the electrodes to be furnished for each type. Samples shall be selected at random from electrodes which have been manufactured using the same production processes, procedures, and equipment which will be used in filling the contract. The samples shall be selected from either filled unit containers or from the production line immediately before packaging. If selected from the production line, the total sample shall be selected throughout the run so all parts of the run are represented.

4.3.1.1 Sampling for visual and dimensional requirements. To verify compliance with visual and dimensional requirements, select test samples in accordance with MIL-STD-105, at inspection level II and acceptable quality level (AQL) = 2.5 percent defective.

4.3.1.2 Sampling for tests for all other requirements. For all other tests, select a random sample in accordance with MIL-STD-105, at inspection level S-3, with an acceptance number of zero.

4.3.2 Sampling for quality conformance inspection.

4.3.2.1 Sampling for examination and test of electrodes. Sample electrodes shall be selected either from filled unit packages or from the production line after the baking operation. The unit packages required for this examination may be selected from the sample of packages used for 4.3.1.1 and 4.3.1.2. If sample electrodes are selected from filled unit packages, the total sample shall be in accordance with MIL-STD-105 for inspection level S-4 with an Acceptable Quality Level (AQL) of 4.0 percent for electrode sizes 1/8 inch (3.25) and larger and an AQL of 6.5 percent for electrode sizes 3/32 inch (2.5) and smaller for single sampling. Lot size shall be expressed in pounds (kg). A minimum number of unit packages shall be selected to permit sampling of five electrodes maximum from each 50-pound (23 kg) package or one electrode from each 10-pound (5 kg) package. If the sample electrodes are selected from the production line, the total sample shall be equivalent to the above and the electrodes shall be selected throughout the production period so that all parts of the lot are represented. The electrodes selected shall be identified as to type, lot, size and other available information such as contract or order number being filled.

4.4 Classification of tests. The inspection and testing of electrodes shall be classified as follows:

a. First article inspection (see 4.5).

b. Quality conformance inspection (see 4.6).

4.5 First article inspection. First article inspection shall consist of tests for all of the requirements of AWS 5.3 and this specification. In

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testing electrodes, the welder shall not use any unusual technique of manipulating the electrode, i.e., any technique not commonly taught to, and proficiently used by, the average welder who has been trained and qualified to use electrodes of the types covered by this specification. The responsibility for performing first article inspection shall be as specified by the procuring activity (see 6.2). If multiple lots are offered in a given shipment, for which first article inspection has been specified by the purchaser, only one lot shall be subject to the first article inspection. The remaining lots shall be subject to quality conformance inspection only. When first article inspection is required, the procurement documents should state:

a. Where the first article inspection is to be performed: at the manufacturer's plant, at a Government laboratory, or at a commercial laboratory.

b. That the approval of first article samples or the waiver of first article inspection shall not relieve the contractor of his obligation to fulfill all other requirements of this specification and of the contract.

4.5.1 Special instructions for first article tests.

4.5.1.1 Information to be furnished by manufacturers in all cases. When submitting samples for first article inspection, or when furnishing the results of such an inspection, the manufacturer shall state in writing:

a. The nominal diameter of the covered electrode coverings (see 1.2).

b. The type classification for which approval is sought (see 1.2).

c. The principal compounds and percentages thereof in the coverings.

d. The range of chemical composition of the deposited weldmetal (see 3.3).

4.5.1.2 Additional information to be furnished when manufacturer performs first article inspection. The manufacturer shall furnish a complete inspection report showing the results of his inspection. The report shall include the following:

a. Results of all tests, referring specifically to the appropriate paragraph numbers of the specification. If a particular requirement is not applicable, the report shall so state it.

b. A complete description of the inspection testing equipment and procedures.

c. Copies of inspection log sheets.

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4.5.1.3 Waiver of first article inspection (see 6.3). First article inspection for a specific type, class, and size of electrode may be waived at the option of the procuring activity when both of the following conditions have been met:

a. A first article specimen of the product has passed all of the first article inspection requirements within the previous four years or the manufacturer is in continuous production of the electrode from contract to contract.

b. The contractor certifies in writing that the chemical composition, dimensions and tolerances, and method of manufacture of the electrode are the same as those of the product which previously passed the first article inspection requirements.

4.6 Quality conformance inspection. Quality conformance inspection shall consist of the examination of 4.6.1 and the test of 4.6.2.

4.6.1 Examination. Sample electrodes, selected in accordance with 4.3.1.1 shall be visually examined for workmanship, and dimensionally measured to determine if the requirements of AWS 5.3 and this specification have been met.

4.6.2 Tests. Sample electrodes, selected in accordance with 4.3.1.2, shall be tested for chemical composition. Welds produced with electrodes shall be tested for tensile strength, and transverse bending for soundness in accordance with methods specified in AWS 5.3.

5. PACKAGING.

5.1 Preservation packaging and packing. Preservation packaging and packing shall be level A or level C (for Army procurement, level C is defined as commercial) of MIL-W-10430, as specified (see 6.2). A unit package shall contain one type, class, and size of electrode. A unit package quantity shall be 1(0.5), 3(1.4), 5(2.3), or 10(5) pounds (kilograms) as specified (see 6.2).

5.2 Marking. In addition to any special markings required by the contract, interior packages, exterior shipping containers, and palletized unit loads shall be marked in accordance with MIL-W-10430.

5.2.1 Warning label. All packages containing covered electrodes, or all individually packaged units enclosed within a larger package shall carry the warning label specified in AWS A5.3. In addition, containers shall bear the following printed label:

CAUTION

The presence of moisture, dirt, grease or other contamination on electrodes can cause weld porosity. Electrodes should be stored in dry, clean locations. Electrodes exposed to moisture should be reconditioned and stored in heated containers.

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5.3 Shipping container tests. Tests on shipping containers specified in MIL-W-10430 shall apply to level A packing only.

6. NOTES.

6.1 Intended use.

6.1.1 MIL-1100. Electrodes of the MIL-1100 classification produce weld metal of high ductility, good electrical conductivity, and a minimum tensile strength of 12,000 psi (90 MPa). MIL-1100 electrodes are used to weld aluminum alloys 1100, 1350 (EC) and other pure aluminum alloys.

6.1.2 MIL-3003. Electrodes of the MIL-3003 classification produce weld metal of high ductility and a minimum tensile strength of 14,000 psi (100 MPa). MIL-3003 electrodes are used to weld aluminum alloys 1100 and 3003.

6.1.3 MIL-4043. The MIL-4043 classification contains approximately five percent silicon, which provides superior fluidity at welding temperatures and for this reason is preferred for general purpose welding. The MIL-4043 classification produces weld metal with fair ductility and minimum tensile strength of 14,000 psi (100 MPa). MIL-4043 electrodes can be used to weld the 6XXX series aluminum alloys, the 5XXX series aluminum alloys (up to 2.5 percent Mg content), aluminum-silicon casting alloys, as well as aluminum alloys 1100, 1350 (EC) and 3003.

6.2 Ordering data. Procurement documents should specify:

6.2.1 Procurement requirements.

a. Title, number and date of specification.
b. Type and size electrodes required (see 1.2).
c. Level of preservation packaging, packing and marking required (level A or level C).
d. Quantity of electrodes required. Electrodes should be ordered by weight in either pounds or kilograms.
e. Type and size of shipping containers.
f. Whether first article inspection (see 4.5) shall be required on a particular contract (see 6.3).
g. Responsibility for performance of first article inspection (see 4.5).

6.2.2 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of the Defense Acquisition Regulatory System (DARS), the data requirements identified below will be developed as specified by an approved Data Item Description DD Form 1664 and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DARS are not invoked, the data specified below may be delivered

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by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph no.	Data requirement title	Applicable DID no.
3.1, 4.3.1	First Article Inspection Procedure First Article Inspection Report	DI-T-4901 DI-T-4902
4.5.1.3	Certificates of Compliance	DI-E-2121

(Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the acquiring activity or as directed by the contracting officer.)

6.3 First article inspection. First article inspection should be kept to a minimum, although it may be required by the purchaser on any contract. When considering whether or not to waive first article inspection, prospective purchasers are advised to consult the Defense General Supply Center, Richmond, Virginia 23297, for information concerning manufacturers whose products have received first article approval in the previous four years under any DoD contract or who are in continuous production of electrodes from one DoD contract to another (see 4.5.1.3). Under these conditions, it is recommended that first article inspection be waived.

6.4 Moisture control of electrodes. The presence of moisture in the electrode covering is a major cause of weld porosity. Dirt, grease or other contamination on the electrode can also contribute to porosity. The absorption of moisture can be quite rapid and the deterioration of the covering can occur after only a few hours exposure to a humid atmosphere. For this reason, unopened packages of electrodes should be stored in a dry, clean location. Electrodes taken from previously opened packages or those exposed to moisture should be "reconditioned" by holding at 350 F to 400 F (177 to 204 C) for an hour before welding. After conditioning, they should be stored in a heated cabinet until use.

6.5 Metric units. The values stated in U.S. customary units are standard. SI values in parentheses are for information only.

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Project No. 3439-A167

☆U.S. GOVERNMENT PRINTING OFFICE: 1980—703-023/164

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